

Properties of Unshunted and Resistively Shunted Nb/AIO_x-Al/Nb Josephson Junctions With Critical Current Densities From 0.1 to 1 mA/ μm^2

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Abstract—We investigated current–voltage characteristics of unshunted and externally shunted Josephson junctions (JJs) with high critical current densities J_c in order to extract their basic parameters and statistical characteristics for JJ modeling in superconducting integrated circuits as well as to assess their potential for future technology nodes. Nb/AIO_x-Al/Nb junctions with diameters from 0.5 to 6 μm were fabricated using a fully planarized process with Mo or MoN_x thin-film shunt resistors with sheet resistance $R_{sq} = 2 \Omega/\text{sq}$ and $R_{sq} = 6 \Omega/\text{sq}$, respectively. We used our current standard MIT Lincoln Laboratory process node SFQ5ee to fabricate JJs with $J_c = 0.1 \text{ mA}/\mu\text{m}^2$ and our new process node SFQ5hs (where “hs” stands for high speed) to make JJs with $J_c = 0.2 \text{ mA}/\mu\text{m}^2$ and higher current densities up to about $1 \text{ mA}/\mu\text{m}^2$. Using *LRC* resonance features on the *I*–*V* characteristics of shunted JJs, we extract the inductance associated with Mo shunt resistors of 1.4 pH/sq. The main part of this inductance, about 1.1 pH/sq, is the inductance of the 40-nm Mo resistor film, while the geometrical inductance of superconducting Nb wiring contributes the rest. We attribute this large inductance to “kinetic” inductance arising from the complex conductivity of a thin normal-metal film in an electromagnetic field with angular frequency ω , $\sigma(\omega) = \sigma_0/(1 + i\omega\tau)$, where σ_0 is the static conductivity and τ the electron scattering time. Using a resonance in a large-area unshunted high- J_c junction excited by a resistively coupled small-area shunted JJ, we extract the Josephson plasma frequency and specific capacitance of high- J_c junctions in 0.1–1 mA/ μm^2 J_c range. We also present data on J_c targeting and JJ critical current spreads. We discuss the potential of using 0.2-mA/ μm^2 JJs in very large scale integration single flux quantum circuits and 0.5-mA/ μm^2 JJs in high-density integrated circuits without shunt resistors.

Index Terms—Josephson device fabrication, Josephson plasma resonance, kinetic inductance, Nb/AIO_x/Nb junctions, SFQ electronics, superconducting device fabrication, superconducting electronics, superconducting electronics fabrication.

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I. INTRODUCTION

SIMPLE superconductor electronic circuits hold records for the highest clock frequency [1] and the lowest energy dissipation per bit operation [2]. The main challenge for this technology has been scalability [3]. Recent developments of the fabrication technology for superconducting Single Flux Quantum (SFQ) circuits at MIT Lincoln Laboratory (MIT LL) [4], [5] have enabled very large scale integration (VLSI) of SFQ test circuits, specifically ac-biased shift registers [6] with 202 kbit and nearly 10^6 Josephson junctions (JJs) [7], approaching the end of scaling predicted in [3] for the technology with one layer of resistively shunted junctions. The current MIT LL technology node SFQ5ee [5] utilizes Nb/AIO_x-Al/Nb Josephson junctions with the Josephson critical current density, J_c of $0.1 \text{ mA}/\mu\text{m}^2$. This process was designed to be used for energy-efficient superconducting multi-bit processors with 10 GHz to 20 GHz clock frequencies for high performance computing as the target application [8].

Increasing the integration scale of SFQ circuits above 10^6 JJs per cm^2 requires reducing the area occupied by Josephson junctions and shunt resistors. This can be achieved by increasing J_c above the current standard of $0.1 \text{ mA}/\mu\text{m}^2$ and going eventually to self-shunted high- J_c junctions requiring no external shunting. A VLSI process with higher- J_c junctions would also provide higher clock frequencies because the maximum clock frequency scales proportionally to the Josephson plasma frequency $f_p = (J_c/2\pi\Phi_0 C_s)^{1/2}$ [9], where C_s is the junction specific capacitance and $\Phi_0 \equiv h/2e$ is the flux quantum.

It has been typically observed that reproducibility and parameter spreads of high- J_c junctions is inferior to low- J_c junctions, which limits the integration scale. The J_c value of $0.1 \text{ mA}/\mu\text{m}^2$ has been viewed as a likely upper limit from the point of view of junction J_c uniformity and reproducibility. However, there are applications, e.g., in digital signal processing, where achieving the highest clock frequency is a more important requirement than increasing the circuit complexity.

In order to address the need for a VLSI process with high- J_c junctions for applications requiring high clock frequencies and to increase the integration scale, we have introduced a new technology node SFQ5hs, where ‘hs’ stands for ‘high speed’. This

node targets higher critical current densities $J_c = 0.2 \text{ mA}/\mu\text{m}^2$ and $0.5 \text{ mA}/\mu\text{m}^2$, but in all other respects is identical to the SFQ5ee node [5].

Designing SFQ circuits with complexities that would make them useful for applications and competitive with semiconductor circuits requires advanced simulation tools and accurate knowledge of the parameters of junctions and other circuit components because of narrow timing margins typical of the SFQ circuits. The properties of unshunted junctions with $J_c = 0.1 \text{ mA}/\mu\text{m}^2$ have been studied due to their use in MIT LL and ADP2 (AIST-Japan) processes, see [4], [5], [10]–[14] and references therein. The available information on the properties of junctions with $J_c = 0.2 \text{ mA}/\mu\text{m}^2$ and their parameter spreads is very limited [9], [15], [16] and almost completely absent for junctions with $J_c = 0.5 \text{ mA}/\mu\text{m}^2$ and higher. In this work we studied electrical properties of unshunted and resistively shunted junctions with J_c from $0.1 \text{ mA}/\mu\text{m}^2$ to about $1 \text{ mA}/\mu\text{m}^2$ in order to extract (or refine) their basic parameters such as plasma frequency, specific capacitance, internal shunting resistance, external shunt inductance, critical current spreads, etc., for use in the junction models for circuit simulations, and to assess the potential of the high- J_c junctions for future technology nodes.

II. EXPERIMENTAL RESULTS

A. Critical Current Density Dependence on Oxygen Exposure

A detailed description of our fully planarized fabrication process and its nodes SFQ4ee and SFQ5ee was given in [4], [5] and also reviewed in [3]. For the SFQ5hs node we keep the same minimum feature size of the critical wiring layers of $0.35 \mu\text{m}$ and the junction minimum diameter of $0.7 \mu\text{m}$, but target higher Josephson critical current density of Nb/ AlO_x /Al/Nb trilayers.

It is generally accepted that J_c of Nb/ AlO_x -Al/Nb junctions is mainly controlled by the oxygen exposure during aluminum oxidation $E = P_{\text{O}_2} \cdot t$, where P_{O_2} is the oxygen partial pressure, and t is the oxidation time. Many results summarized in [17]–[19] suggested the existence of a universal $J_c(E)$ proportional to $E^{-\alpha}$ dependence with the exponent changing from $\alpha \sim 0.4$ in the low- J_c range to $\alpha \sim 1.6$ in the high- J_c range at about $0.1 \text{ mA}/\mu\text{m}^2$. The latter, very steep, $J_c(E)$ dependence above $0.1 \text{ mA}/\mu\text{m}^2$ has been viewed as a serious challenge for a reproducible high- J_c junction fabrication process.

In order to further examine the dependence of J_c on P_{O_2} and t , we used dynamic oxidation of the aluminum layer and investigated several oxidation regimes: in pure oxygen; in O_2 -Ar mixture with 3% of oxygen; and in O_2 -He mixture with 3% of oxygen. Trilayer depositions were done in an Endura 5500 PVD cluster tool (Applied Materials, Inc.) using magnetron sputtering. Aluminum oxidation was done in a reactive chamber with a turbo-molecular pump.

The results are presented in Fig. 1. It also includes our results for the low- J_c region accumulated over several years by using oxidation in pure oxygen in different chambers of the Endura cluster and in a second deposition system, CVC Connexion cluster tool. For a comparison we also show the results of

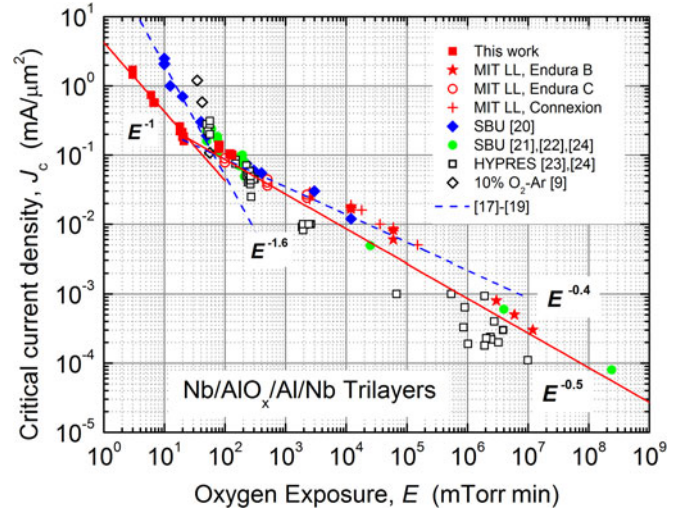


Fig. 1. Critical current density of Nb/ AlO_x -Al/Nb junctions as a function of oxygen exposure during AlO_x tunnel barrier formation by oxidation of the aluminum layer at room temperature. Historical data from [17]–[19] are shown by the blue dashed lines, corresponding to $J_c \propto E^{-1.6}$ and $E^{-0.4}$ dependences. More recent data shown by blue diamonds (◆) and green circles (●) are from the Stony Brook University group [20]–[22], [24], by open squares (□) and diamonds (◇) are from HYPRES fab [9], [23], [24]. MIT-LL data are shown by the red symbols (■), (*), (○), and (+). Our high- J_c data are well described by $J_c \propto E^{-1}$ dependence (red solid line).

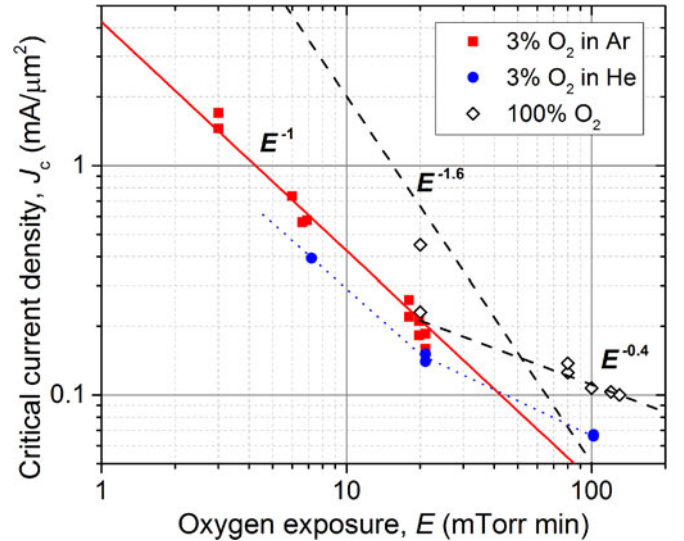


Fig. 2. High- J_c region of the critical current density dependence on oxygen exposure $E = P_{\text{O}_2} t$ during AlO_x barrier formation in Nb/ AlO_x -Al/Nb junctions. The solid line is $J_c \propto E^{-\alpha}$ dependence with $\alpha = 1$; the dashed lines correspond to $\alpha = 1.6$ and $\alpha = 0.4$ [17], [18]. The dotted line connecting the data points for oxidation in 3% O_2 -He mixture is to guide the eye. The junctions were fabricated by a fully planarized process with four Nb layers. The J_c was obtained from 4.2-K measurements of the critical current I_c of the junctions with design diameters from 0.7 to $2.2 \mu\text{m}$.

the Stony Brook University group [20]–[22] obtained in 1996–2002 and of the HYPRES foundry [9], [15], [23], [24] obtained during 2003–2012 period. Superimposed are the $J_c \propto E^{-\alpha}$ dependences shown by blue dashed lines marked $E^{-0.4}$ and $E^{-1.6}$, fitting the data obtained in [17], [18] prior to 1995.

More detailed results for the high- J_c region, obtained with the SFQ5hs process using oxidations in pure O₂, O₂-Ar, and O₂-He mixtures are shown in Fig. 2. We obtained $\alpha = 1$ in the high- J_c region, a much weaker $J_c(E)$ dependence than the previously reported $\alpha = 1.6$, indicating that the exponent in $J_c \propto E^{-\alpha}$ dependence is not universal but rather depends strongly on the oxidation conditions. It also appears that the crossover into the high- J_c regime in our trilayers occurs at a much lower exposure of about 20 mTorr·min and at a higher J_c of about 0.2 mA/ μm^2 rather than at 0.1 mA/ μm^2 considered to be the boundary of the low- J_c regime in the prior publications.

Comparing oxidations in pure O₂ and O₂-Ar mixtures with different dilution, we concluded that, at the same E , a longer oxidation at lower O₂ partial pressure (higher Ar dilution) results in a lower J_c and higher-quality junctions than shorter oxidations at high pressures. Helium dilution resulted in the lowest J_c at the same nominal oxygen exposure. It might be interesting therefore to evaluate O₂-Kr mixtures for barrier oxidation process.

Oxygen exposures at 20 mTorr·min and below done in pure oxygen at 4 mTorr and lower pressures did not give us reproducible J_c results. However, reproducible results at $J_c = 0.2 \text{ mA}/\mu\text{m}^2$ were obtained by using at least 30-min-long oxidations in diluted O₂ at pressures of about 20 mTorr, thus making our SFQ5hs process with 0.2-mA/ μm^2 JJs a stable technology node.

We note in this respect that $J_c \propto E^{-0.5}$ dependence typical for low- J_c junctions was observed in [25] up to J_c about 0.47 mA/ μm^2 , using 10-min oxidations in pure O₂. However, this result has not been reproduced since then. We expect that $J_c = 0.5 \text{ mA}/\mu\text{m}^2$ could be also a stable technology node based on the fabrication runs produced so far. We also foresee evaluation of oxidations in 1% O₂ in Ar, similarly to [10].

The increase of the oxidation exponent α in the high- J_c region is usually attributed to the appearance of defects ('pin-holes,' quantum point contacts) in the tunnel barrier, which in the superconducting state give additional channels of the subgap current transport via multiple Andreev reflections; see [26]–[28], and references therein. These defects are likely just oxygen vacancies creating atomic-size high-transparency regions in the barrier [29], [30]. We suggest that the number of these nonequilibrium defects decreases by conducting the oxidations at higher dilutions, thus maintaining a high overall gas pressure and giving oxygen more time to fill in all the required positions. The overall pressure controls the amount of collisions between the gas molecules and the wafer surface, providing momentum transfer from Ar atoms to oxygen atoms on the surface, increasing their surface mobility.

B. I - V Characteristics of Junctions, $J_c = 0.1 \text{ mA}/\mu\text{m}^2$

In order to evaluate the transport properties, we measured current-voltage (I - V) characteristics of junctions with design diameters from 0.7 μm to 2.2 μm at 4.2 K, using unshunted and resistively shunted junctions. The typical I - V characteristics of the unshunted tunnel junctions with

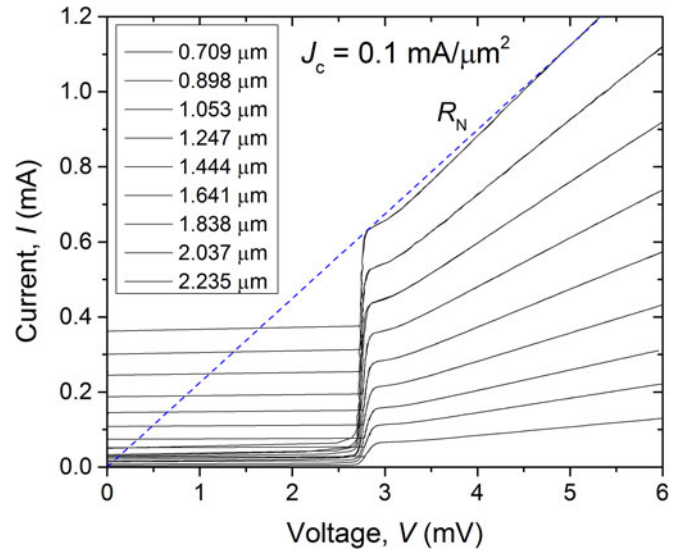


Fig. 3. The typical current-voltage characteristics of Nb/AlO_x-Al/Nb tunnel junctions with $J_c = 0.1 \text{ mA}/\mu\text{m}^2$. Design diameter of the circular junctions used is shown in the legend, corresponding to the curves from bottom to top. The gap voltage of the junctions is 2.8 mV and $R_{sg}/R_N = 10$.

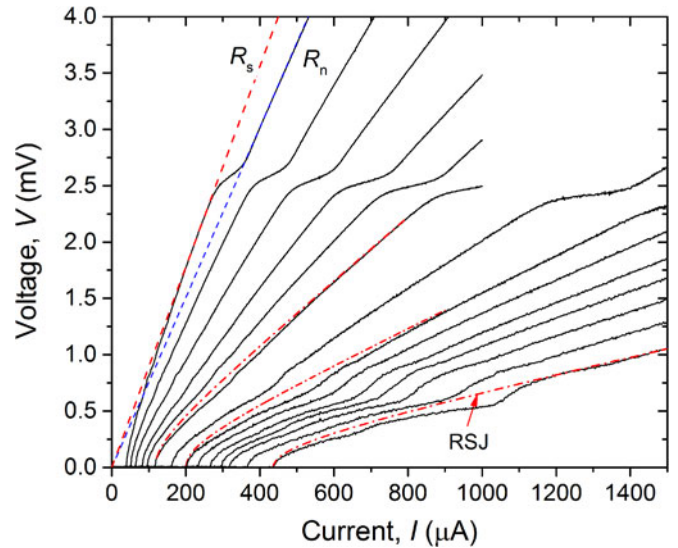


Fig. 4. Current-voltage characteristics of resistively shunted junctions at $\beta_c = 0.2$ and $J_c = 0.1 \text{ mA}/\mu\text{m}^2$. The black curves from the top to bottom (and from the left to right) correspond to junctions with design diameters (in micrometers) of 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.4, 1.6, 1.7, 1.8, 1.9, 2.0, 2.14, and 2.34, respectively. The red and blue dashed lines are the linear approximations defining (1) and (2) in the subgap and above the gap voltage regions, respectively. Dashed-dotted red curves show the RSJ ($\beta_c \ll 1$) model dependence $V = R_s(I^2 - I_c^2)^{1/2}$ for the junctions with 1.2-, 1.6-, and 2.34- μm diameters.

$J_c = 0.1 \text{ mA}/\mu\text{m}^2$ fabricated by the 8-layer niobium process SFQ4ee are shown in Fig. 3.

We varied the shunt resistor in a wide range in order to achieve values of the McCumber-Stewart parameter [31], [32] $\beta_c = 2\pi I_c R_s^2 C / \Phi_0$ from $\beta_c \sim 0.1$ to $\beta_c \gg 1$, where R_s is the effective damping resistance. Fig. 4 shows the typical I - V characteristics of the shunted junctions with $J_c = 0.1 \text{ mA}/\mu\text{m}^2$

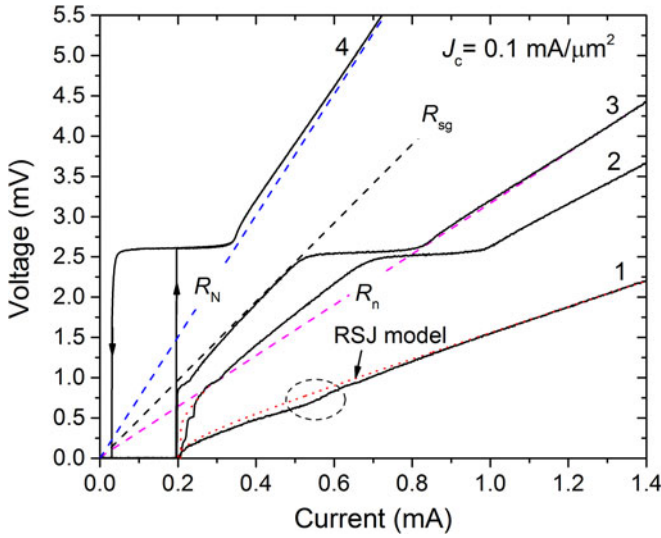


Fig. 5. Current-voltage characteristics of resistively shunted tunnel junctions of $1.6 \mu\text{m}$ diameter and $J_c = 0.1 \text{ mA}/\mu\text{m}^2$. The design parameters of the shunt resistors were selected to obtain the following values of β_c and $V_c \equiv I_c R_s$: “1”— $R_{sh} = 1.6 \Omega$, $\beta_c = 0.2$, and $V_c = 0.30 \text{ mV}$; “2”— $R_{sh} = 3.73 \Omega$, $\beta_c = 1.0$, and $V_c = 0.69 \text{ mV}$; “3”— $R_{sh} = 5.4 \Omega$, $\beta_c = 2.0$, and $V_c = 0.96 \text{ mV}$; “4”—unshunted junction. The dotted curves show the RSJ model ($\beta_c \ll 1$) dependence $V = R_s(I^2 - I_c^2)^{1/2}$ for “1” and resistively and capacitively shunted junction (RCSJ) model with $\beta_c = 1.0$ for “2.” Strong deviations from the RSJ model are marked in curve “1.” Note also a voltage plateau at $\sim 0.5 \text{ mV}$ in curve “2.” These features arise from a contribution of shunt inductance; see text.

and shunts designed to target the characteristic voltage $V_c = 0.3 \text{ mV}$ and $\beta_c = 0.2$, where $V_c \equiv I_c R_s$.

In the standard junction model, it is assumed that the internal junction damping by a nonlinear (voltage-dependent) junction resistance can be replaced in the first approximation by a linear resistor R_{sg} in the subgap region of voltages $V < V_g$, and by the junction normal resistance R_N at voltages $V > V_g$, where $V_g = 2\Delta/e$ is the gap voltage, Δ is the superconducting energy gap in Nb electrodes. With an external shunt resistor, R_{sh} in parallel with the internal resistance, the effective shunting resistance in the subgap region becomes

$$R_s = R_{sh} R_{sg} / (R_{sh} + R_{sg}), \quad V < V_g \quad (1a)$$

and in the ‘normal’ resistance region

$$R_n = R_{sh} R_N / (R_{sh} + R_N), \quad V > V_g \quad (1b)$$

The shunt resistor realized on the fabricated wafers, R_{sh} may slightly differ from the design value, R_{shd} , resulting in $R_{sh} = k R_{shd}$ with $k \approx 1$, because the sheet resistance of the resistor films, may slightly deviate from the target values of $2 \Omega/\text{sq}$ and $6 \Omega/\text{sq}$, respectively.

Fig. 5 shows the typical I - V characteristics of the same-size junctions with different shunt resistors. At small resistance values ($\beta_c < 0.4$), the effective shunt resistance R_s can be extracted by fitting the I - V characteristic to the resistively shunted junction (RSJ) model $V = R_s(I^2 - I_c^2)^{1/2}$. At larger β_c we used the slope of the I - V curves as shown in Figs. 4 and 5. Similarly, R_n was obtained from the slopes of the linear dependences at $V > V_g$. The effective shunt resistances obtained in this

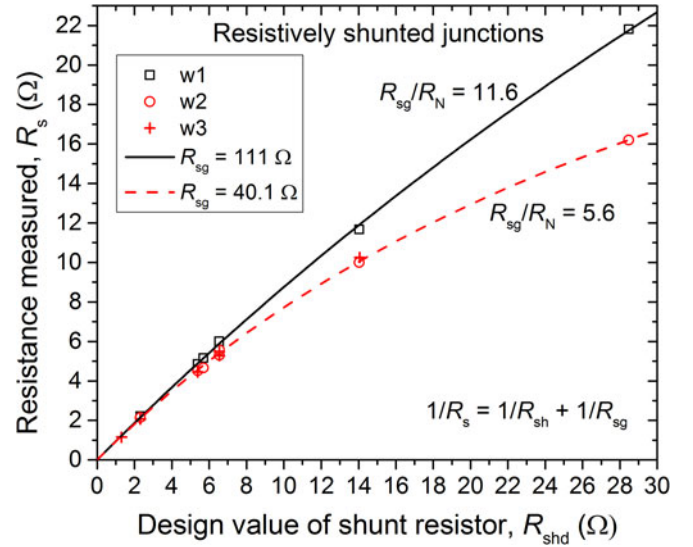


Fig. 6. Extracted values of the effective shunting resistance, R_s in the subgap region as a function of design parameters of the shunt resistor, R_{shd} . Resistively shunted Josephson junctions with diameter of $1.6 \mu\text{m}$ were used. Solid and dashed curves show (1a) at two values of the subgap resistance, R_{sg} : 111Ω , representing wafer 1 (w1), and 40.1Ω , representing w2 and w3. The R_{sg} and R_N were measured independently using unshunted junctions on the same chip; $k = 0.95$ in $R_{sh} = k R_{shd}$ was determined from the sheet resistance measurements of the resistor film on the wafers at 4.2 K .

manner are shown in Fig. 6 as a function of the design shunt resistor value for three wafers with different R_{sg}/R_N ratios characterizing the tunnel barrier quality (subgap leakage). The R_{sg} and R_N were measured independently using unshunted junctions, and k was also obtained from independent measurements of the sheet resistance and the resistor width bias.

As can be seen in Fig. 6, (1a) holds with a good accuracy in the entire range of resistor values studied. An excellent agreement was also obtained between the measured values of R_n and (1b), giving a good self-consistency check.

Fig. 6 also stresses the importance of the subgap resistance in high- J_c junctions, which can almost always be neglected in shunted low- J_c junctions with $R_{sg}/R_N \gg 1$. In our $J_c = 0.1 \text{ mA}/\mu\text{m}^2$ process, the typical value is $R_{sg}/R_N \approx 11$, but higher subgap leakage wafers are frequently observed with R_{sg}/R_N going down to about 6.

Even smaller R_{sg}/R_N ratios, in the range from 3 to 6, are observed for junctions with $J_c = 0.2 \text{ mA}/\mu\text{m}^2$, see Fig. 7. Subgap resistance variations may cause variability of the shunting resistance and the voltage V_c characterizing the junction switching speed. A simple way to reduce this variability is to use a lower shunt resistor value, over-shunt junctions, making the relative contribution of R_{sg} smaller. Over-shunting junctions reduces somewhat the maximum clock frequency but makes circuits less sensitive to the fabrication process variations.

The I - V characteristics of resistively shunted JJs in Fig. 5 display a few interesting features. At small β_c values, the I - V curves deviate downward from the RSJ model (e.g., curve 1 in Fig. 5) at low voltages and have an upward bend and a peak in the differential resistance dV/dI in the region marked in Fig. 5. At $\beta_c = 1$, the value used most often in SFQ circuit design,

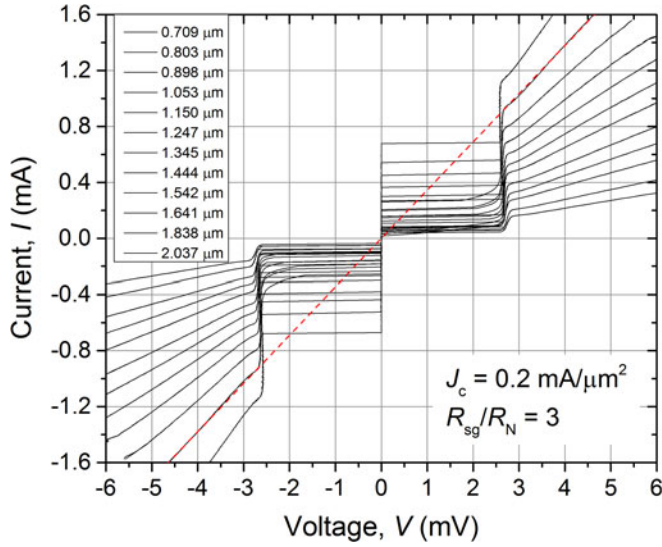


Fig. 7. Current-voltage characteristics of Nb/AlO_x/Al/Nb tunnel junctions with $J_c = 0.2 \text{ mA}/\mu\text{m}^2$. The curves from the top to bottom on the left side correspond to the junctions with design diameters given in the legend. The linear part of the I - V characteristics at $V > V_g$ extrapolates to zero as shown by the dashed line, indicating that there is no 'excess' current.

the I - V has a plateau at $V \sim 0.5 \text{ mV}$ and less sharp features at $\sim 0.25 \text{ mV}$ and $\sim 1 \text{ mV}$. These features do not follow from the pure RSJ or from resistively and capacitively shunted junction (RCSJ) models but indicate the importance of the inductance associated with the resistive shunt, the so-called RLCSJ model [32]–[36], see Section III.

C. I - V Characteristics of Junctions, $J_c = 0.2 \text{ mA}/\mu\text{m}^2$

The typical I - V characteristics of tunnel junctions with $J_c = 0.2 \text{ mA}/\mu\text{m}^2$ are shown in Fig. 7. The junctions are still highly hysteretic, although they have lower R_{sg}/R_N ratios than $0.1 \text{ mA}/\mu\text{m}^2$ junctions, and hence require external shunting for applications in digital circuits. There is no 'excess' current; the linear part of the I - V characteristics at $V \gg V_g$ extrapolates to zero current, as shown by the dashed line in Fig. 7.

The I - V characteristics of the shunted junctions with $1.6\text{-}\mu\text{m}$ diameter are shown in Fig. 8. We used exactly the same design and values of the shunt resistors as for the $0.1\text{-mA}/\mu\text{m}^2$ junctions. Therefore, the β_c parameter increased proportionally to the J_c , and additionally because of an increase in the junction specific capacitance C_s , overall slightly more than by a factor of 2. Note, that, with increasing J_c , some features in the I - V curves changed significantly. For instance, a deviation from the RSJ model in Fig. 5 curve 1, became a sharp feature at the same value of R_{sh} , shown by a dashed circle in Fig. 8.

Also, the dc voltage across the junction (curve 1) is lower than in the simple RSJ model in the entire range of currents. This also implies that the amplitude of SFQ pulses generated by the junction is lower than in the RSJ model.

D. I - V Characteristics of Junctions, $J_c = 0.8 \text{ mA}/\mu\text{m}^2$

Fig. 9 shows I - V characteristics of junctions with yet higher $J_c \approx 0.77 \text{ mA}/\mu\text{m}^2$. The junctions' subgap resistance is about

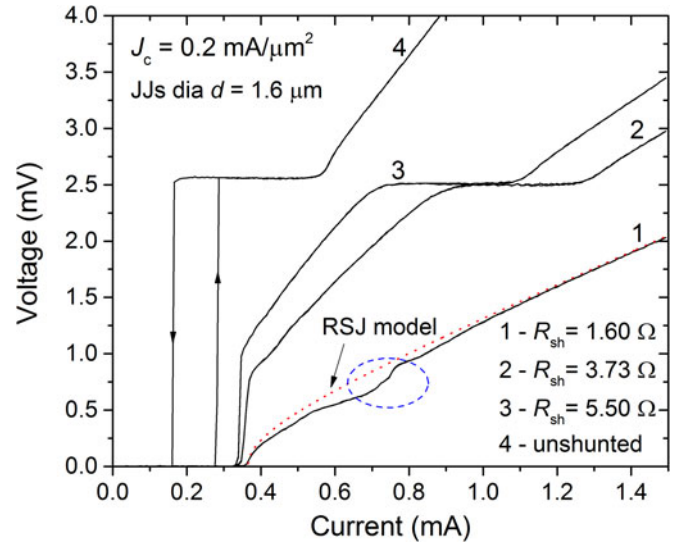


Fig. 8. Current-voltage characteristics of the resistively shunted Nb/AlO_x-Al/Nb tunnel junctions with $J_c = 0.2 \text{ mA}/\mu\text{m}^2$. The design diameter of the junctions is $1.6 \mu\text{m}$. The shunt resistors are exactly the same as used for $0.1\text{-mA}/\mu\text{m}^2$ junctions in Fig. 5. The dotted line is the RSJ model ($\beta_c \ll 1$); the region of strong deviations from the RSJ model is circled.

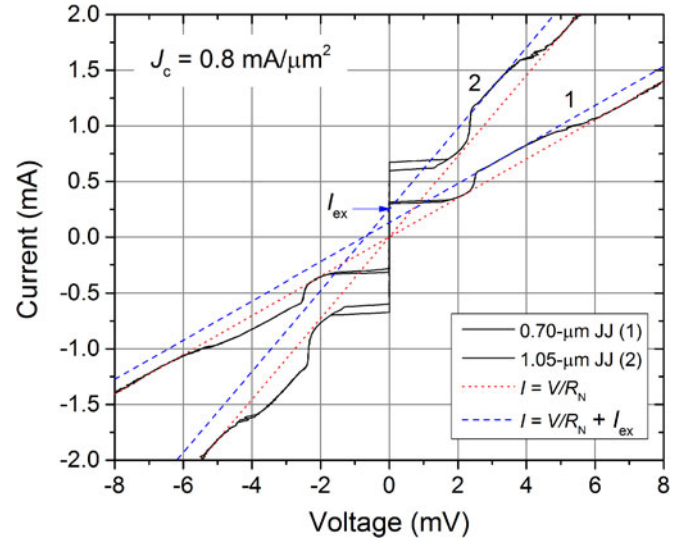


Fig. 9. The typical I - V characteristics of Nb/AlO_x-Al/Nb tunnel junctions with $J_c = 0.77 \text{ mA}/\mu\text{m}^2$. The junctions become almost completely self-shunted with $I_r/I_c \approx 0.87$. Dashed lines show the presence of the 'excess' current $I_{ex}/I_c \approx 0.36$. Dotted lines show the normal-state tunneling characteristics.

the same as R_N as shown by the dotted lines. The return (retrotrapping) current, I_r is about $0.87I_c$. There is a large 'excess' current at $V > V_g$, i.e., the linear part of the I - V characteristics does not extrapolate to zero and can be described by $I = V/R_N + I_{ex}$, where I_{ex} is the excess current, $I_{ex} \approx 0.36I_c$. At large voltages the excess current diminishes, and the simple dependence $I = V/R_N$ recovers. Diminishing of the excess current is likely caused by an increase in the junction internal temperature due to Joule heating. Indeed, heating and/or nonequilibrium quasiparticle effects are indicated by a progressive decrease of the apparent gap voltage in junctions with increasing of the internal

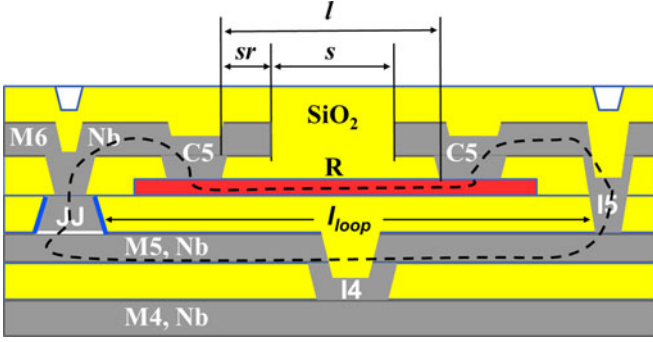


Fig. 10. A schematic cross section of a resistively shunted junction, showing a current loop (dashed black curve) associated with the shunt resistor R . The loop is formed by the junction bottom electrode M5 with width w_{M5} , top wire M6 with width w_{M6} and the resistor. The vias to the junction, resistor, and between superconducting layers are marked C5, I5, and I4, respectively. The loop length, l_{loop} is the distance between the JJ and I5 via. In all resistor designs we used $w_{M5} = w_{M6}$.

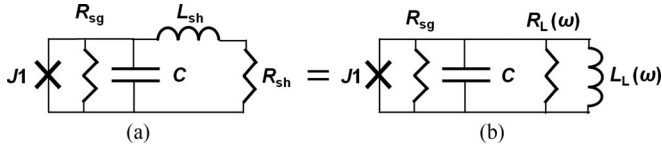


Fig. 11. (a) Shunted junction circuit diagram taking into account inductance L_{sh} associated with the resistive shunt R_{sh} and the junction capacitance C . (b) Series inductance L_{sh} creates a frequency-dependent damping $R_L(\omega)$ and inductive load $L_L(\omega)$.

heat power $I_c V_g$. Also, a significant noise appears during I - V measurements at currents above about 1.6 mA, also indicating heating effects.

It is interesting to note that the junctions switch back into the superconducting state right below a feature corresponding to a single Andreev reflection process at $V = \Delta/e$. This indicates that the subgap conductance by higher-order multiple Andreev reflection processes does not provide sufficient damping to keep the junctions nonhysteretic at this J_c . Nonhysteretic I - V characteristics were obtained at $J_c > 1 \text{ mA}/\mu\text{m}^2$, not presented here.

III. SHUNT INDUCTANCE EFFECTS

A. Resistively, Inductively, and Capacitively Shunted Junction

Figs. 4–6 and 8 show that the I - V characteristics of resistively shunted high- J_c junctions at large voltages are consistent with the simple linear shunt resistance approximation (1). However, a closer inspection reveals many features of the I - V characteristics which do not follow from the resistively shunted junction (RSJ) model, namely voltage plateaus and current steps, which can be seen in Figs. 4, 5, and 8. It was shown in [35]–[38] that this complex behavior is caused by the shunt inductance presumably associated with the geometrical (magnetic) inductance of the loop formed between the junction top and bottom electrodes through the resistor, as shown schematically in Fig. 10. This inductance forms a damped parallel LRC resonator with the junction capacitance, causing various self-induced features on I - V characteristics due to interaction of nonlinear

Josephson oscillations with this resonator. We note in this respect a considerable recent interest in studying self-induced resonance features on I - V characteristics of underdamped JJs and arrays of JJs coupled to series LC resonators; see, e.g., [50]–[52] and references therein.

Due to inductance associated with resistive shunting, the shunted junction should be treated as resistively, inductively and capacitively shunted junction (RLCSJ) with frequency-dependent damping. Its circuit diagram is shown in Fig. 11.

The junction $J1$ with its subgap resistance R_{sg} can be viewed as an rf current source with a finite internal resistance. The impedance seen from this source is

$$Z(\omega) = \frac{R_{sh} + i\omega L_{sh}}{1 - \omega^2 L_{sh} C + i\omega R_{sh} C} \quad (2)$$

Its imaginary part becomes zero at a resonance frequency [33]

$$\omega_r^2 = \frac{1}{L_{sh} C} - \left(\frac{R_{sh}}{L_{sh}} \right)^2 = \omega_0^2 - \omega_L^2, \quad (3)$$

where $\omega_0^2 = (L_{sh} C)^{-1}$ is the L - C resonance frequency in the absence of dissipation and $\omega_L = R_{sh}/L_{sh}$.

The resonance exists if $\omega_0 \geq \omega_L$, that requires $(L_{sh}/C)^{1/2} \geq R_{sh}$, i.e., the resonator impedance be larger than R_{sh} . This is equivalent to the requirement that the resonator be underdamped at frequency ω_L , i.e., $\omega_L \tau_c \leq 1$, where $\tau_c = R_{sh} C$. The corresponding resonance voltage is given by the Josephson relation

$$V_r = \omega_r \Phi_0 / (2\pi). \quad (4)$$

At $\omega = \omega_r$, the impedance seen from the junction is real

$$Z(\omega_r) / R_{sh} = (\omega_L \tau_c)^{-1} \quad (5)$$

and larger than R_{sh} because $\omega_L \tau_c < 1$. Therefore, an increase in the slope of the I - V characteristic, dV/dI occurs near the resonance voltage.

However, the real part of impedance (2) reaches its maximum (has a ‘resonance’) at a higher frequency (voltage), given by

$$\omega_m^2 = \frac{1}{L_{sh} C} - \frac{1}{2} \left(\frac{R_{sh}}{L_{sh}} \right)^2 = \omega_0^2 - \frac{1}{2} \omega_L^2, \quad (6a)$$

$$V_m = \omega_m \Phi_0 / (2\pi), \quad (6b)$$

independently of whether ω_r exists or not. The maximum value of the real part of the impedance is

$$\text{Re} Z(\omega_m) / R_{sh} = \left[(\omega_L \tau_c) - \frac{1}{4} (\omega_L \tau_c)^2 \right]^{-1}. \quad (7)$$

If we normalize voltages to $I_c R_{sh}$ product, $v = V/(I_c R_{sh})$, and introduce dimensionless shunt parameters $\beta_L = 2\pi I_c L_{sh} / \Phi_0$ and $\beta_C = 2\pi I_c R_{sh}^2 C / \Phi_0$, the resonance frequencies can be expressed as

$$v_r = \beta_L^{-1} \left(\frac{\beta_L}{\beta_C} - 1 \right)^{\frac{1}{2}} \quad (8a)$$

$$v_m = \beta_L^{-1} \left(\frac{\beta_L}{\beta_C} - 1/2 \right)^{1/2}, \quad (8b)$$

TABLE I
EXTRACTED INDUCTANCE OF MO SHUNT RESISTORS WITH $R_{sq} = 2 \Omega/\text{sq}$

JJ diameter (μm)	Shunt resistor (Ω)	Resistor width, w (μm)	dV/dI peak voltage (mV)	L_{sh}^+ (pH)	L_{sh}^- (pH)
0.7	9.7	1	0.762	6.60	1.33
0.8	7.1	1	0.744	5.21	0.95
0.9	5.5	1	0.790	3.46	0.76
1.0	4.3	1	0.752	3.15	0.56
1.1	3.5	1	0.798	2.23	0.47
1.2	2.9	1	0.798	1.86	0.38
1.4	2.08	1.2	0.798	1.36	0.27
1.6	1.6	1.5	0.772	1.11	0.21
1.7	1.4	2.0	0.760	1.03	0.18
1.8	1.25	2.0	0.744	0.960	0.16
1.9	1.13	2.0	0.722	0.920	0.14
2.0	1.0	2.1	0.710	0.868	0.12
2.335	0.733	3	0.660	0.749	0.09

and the resonance existence condition as $\beta_L \geq \beta_C$.

B. Extraction of Shunt Inductance, L_{sh}

A good description of all complex features in the I - V characteristics and the value of L_{sh} can be obtained from fitting the characteristics to the RLCSJ model [36]–[38]. For SFQ circuit design it is preferable to not have internal shunt resonances as they complicate the junction dynamics. Therefore, below we will estimate the relevant parameters and the effects of the shunt design based on the position of peaks in dV/dI versus V dependences, leaving a more complete treatment for a separate publication.

The I - V characteristics of $0.1\text{-mA}/\mu\text{m}^2$ JJs shown in Fig. 4 correspond to $\beta_C \approx 0.22$ and $I_c R_{sh} \approx 0.33$ mV. They all display small and relatively broad peaks in $R_{sh}^{-1} dV/dI$ in the following ranges of voltages: 0.2–0.25 mV; 0.45–0.54 mV; 0.64–0.67 mV; 0.74–0.79 mV; 0.85–0.95 mV. The voltage of the strongest peak, which occurs at V_m , is given in Table I. For all JJ sizes, the amplitude of the strongest peak in $R_{sh}^{-1} dV/dI$ is in the range from 1.4 to 2, increasing with the junction size.

The second strongest peak is in the range from 0.2 mV to 0.25 mV, corresponding to the third subharmonic of $V_r \sim 0.67$ mV. The subharmonic features appear because at voltages $V < V_c$ the Josephson oscillations are rich in higher harmonics of the fundamental Josephson frequency $f_J = V/\Phi_0$, so the resonance at V_r can be pumped at the subharmonics of V_r : $V_r/2$, $V_r/3$, etc. At this shunting, however, the second subharmonic feature is small because $V_r/3 < V_c < V_r/2$ and Josephson oscillations at $V > V_c$ are almost purely sinusoidal.

The shunt inductance can be estimated from (6) using the voltage corresponding to the main peak in dV/dI and capacitance $C = C_s A_J$, where A_J is the junction actual area [4] and C_s the specific capacitance. For $0.1\text{-mA}/\mu\text{m}^2$ junctions, we used $C_s = 70\text{ fF}/\mu\text{m}^2$, see Section IV and [40]. Both (3) and (6) are quadratic and have two solutions for L_{sh} , i.e., the same resonance frequency can be obtained using two very different

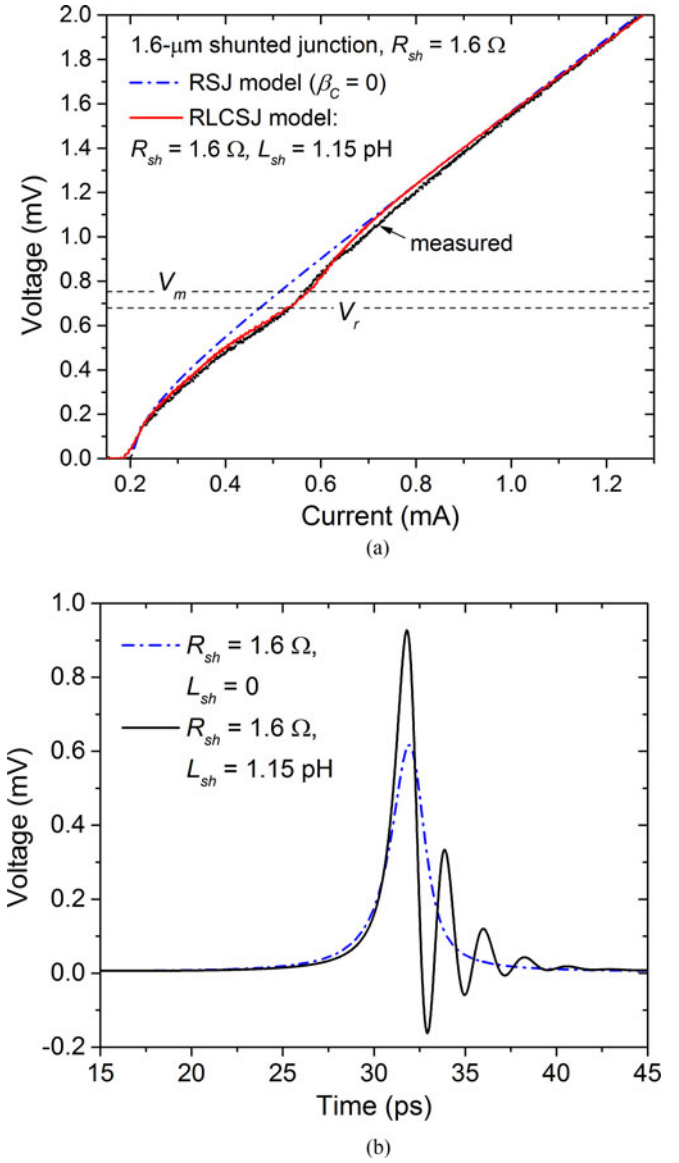


Fig. 12. (a) The measured I - V characteristic of a $1.6\text{-}\mu\text{m}$ -diameter JJ shunted with a $1.6\text{-}\Omega$ molybdenum-film shunt resistor and the simulated characteristics: the RSJ model ($\beta_C = \beta_L = 0$)—blue dashed-dotted curve; the RLCSJ model with the shunt inductance $L_{sh} = 1.15$ pH ($\beta_C = 0.22$, $\beta_L = 0.67$)—solid red curve. The values of the resonance voltage V_r (4) and V_m (6b) at these parameters are shown by the horizontal dashed lines. (b) The simulated voltage waveform (SFQ pulse shape) for a $1.6\text{-}\mu\text{m}$ -diameter JJ with shunt inductance taken into account (black solid curve) and without the shunt inductance (blue dashed-dotted curve).

inductors—a unique feature of LRC resonators. Table I summarizes parameters of molybdenum resistors with $2 \Omega/\text{sq}$ sheet resistance used to shunt the JJs in Fig. 4, the voltage of the dV/dI main peak, and the two solutions for the shunt inductance L_{sh}^+ and L_{sh}^- obtained.

L_{sh}^- is an extraneous solution because, in all cases, it corresponds to a resonator with quality factor $Q < 1$, where $Q = \sqrt{L_{sh}^-}/R_{sh}$. This resonance would not be observable. On the other hand, L_{sh}^+ gives Q in the range from 1.47 to 2, which is consistent with the observed $R_{sh}^{-1} dV/dI$ peak amplitudes in

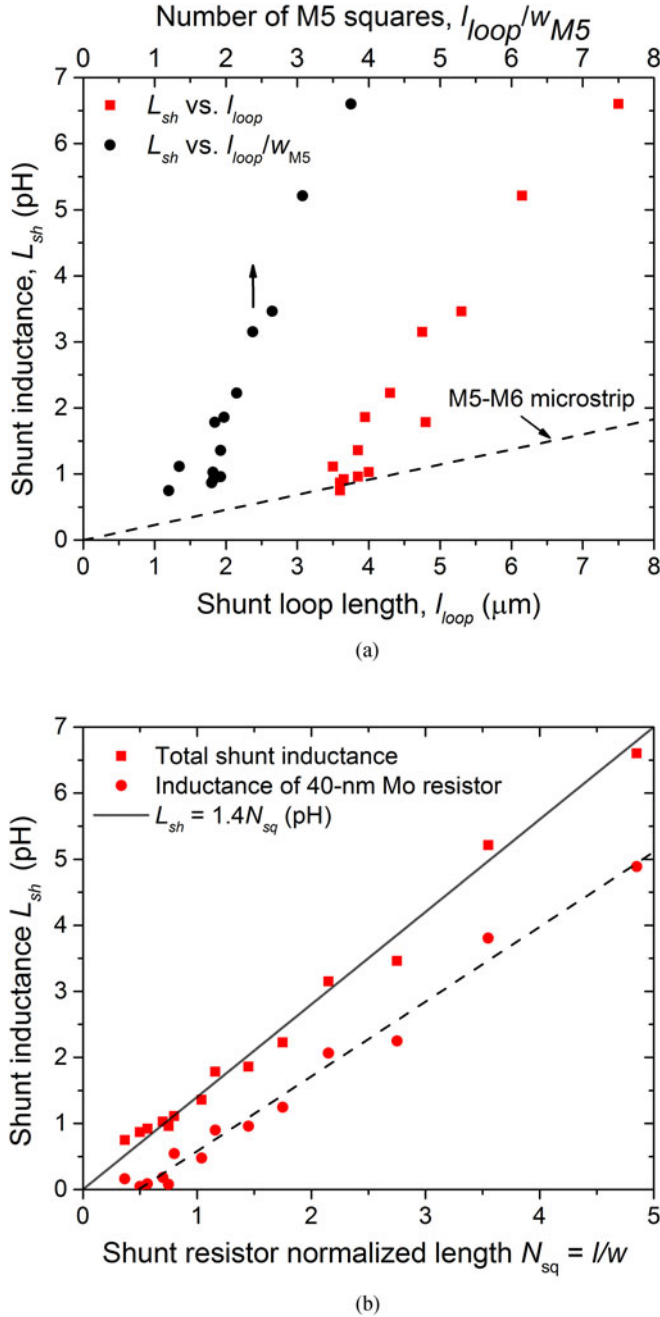


Fig. 13. Shunt inductance, L_{sh} versus Mo shunt resistor design parameters. (a) L_{sh} versus the shunt loop length l_{loop} (bottom axis) and the number of squares l_{loop}/w_{M5} in the bottom electrode (top axis); dashed line is (9) with $\ell = 0.229 \text{ pH}/\mu\text{m}$ for the M5-M6 wiring pair [39], [40]. (b) L_{sh} versus the number of square of the resistor film $N_{sq} = l/w$: ■ – total inductance L_{sh} (resistor + wiring); • – contribution of the resistor film alone, $L = L_{sh} - L_g$. The solid line in (b) is the linear fit of the total inductance L_{sh} versus N_{sq} of the resistor film, giving $L_{sh} = 1.4 N_{sq}$ (in pH). The dashed line is the linear fit to the inductance of the Mo resistor film alone, giving inductance per square, $L_{sq} = 1.132 \text{ pH/sq}$.

the range from 1.44 to 2. Therefore, we will use hereafter L_{sh}^+ for the total shunt inductance.

As an additional verification, we simulated the I - V characteristics of a shunted JJs using L_{sh}^+ in series with the shunt resistor and compared it with the measurements in Fig. 12(a). We see

that the simulated I - V agrees very well with the measured one and captures all main features of the measured characteristics: overall reduction in the time-averaged voltage with respect to the RSJ model; voltage upturn (increase in dV/dI) at $V \approx V_r$; and the change in curvature of $V(I)$, i.e., a maximum in dV/dI , at $V \approx V_m$. Fig. 12(b) shows the simulated voltage waveforms at a current slightly above the I_c in the case of a purely resistive shunt (dashed-dotted curve) and with the shunt inductance taken into account (solid curve).

The shunt inductance is usually assumed to be associated with geometrical inductance of the current loop shown in Fig. 10, [33]–[38]. In this case, the loop inductance is expected to be proportional to the loop length, l_{loop} , the distance from the junction edge to the I5 via along the JJ base electrode, or to the number of squares l_{loop}/w_{M5}

$$L_g = l_{loop}\ell + L_{via}, \quad (9)$$

where ℓ is the inductance per unit length of the microstrip formed by M5 and M6 layers, L_{via} is the total contribution of all vias to the JJ and the shunt resistor, which is independent of the resistor length, and w_{M5} the wire width. Fig. 13(a) shows the dependence of the extracted value L_{sh}^+ on the loop length.

The expected geometrical inductance (9) is shown in Fig. 13(a) by the dashed line. It was calculated using $0.229 \text{ pH}/\mu\text{m}$ for M5-M6 microstrips with widths $w_{M5} = w_{M6} = 2 \mu\text{m}$, which follows from the SQUID-based inductance measurements in [39], [40] and inductance simulations. We can see that the extracted inductance is much larger than the geometrical inductance of the loop and does not scale well with loop length or the number of squares l_{loop}/w_{M5} . The inductance seems to extrapolate to zero at a finite length of the loop, which makes no physical sense. However, the extracted L_{sh} scales linearly with the number of squares (normalized length) l/w in the resistor film, as shown in Fig. 13(b).

In order to get the inductive contribution of the thin normal-metal film alone, we subtracted the contribution of the M5-M6 microstrip from the extracted values of L_{sh} . The result is shown in Fig. 13(b) by solid circles. There are two potential reasons why the resultant dependence extrapolates to zero at a finite length of the resistor. The first reason is that the geometric inductance of the loop is somewhat overestimated because the length of the M6 wire $l_{loop} - l$ is shorter than the loop length, see Fig. 10. The second reason is a weak capacitive coupling between M6 wires and the resistor in a $0.35\text{-}\mu\text{m}$ overlap (surround) between the wires and the resistor between C5 vias, labeled 'sr' in Fig. 10.

C. 'Kinetic' Inductance of Thin-Film Shunt Resistors

It is clear from Fig. 13(b) that the inductance of the resistor film itself, $L = L_{sh}^+ - L_g$ is the dominant contribution to the total shunt inductance, not the inductance of superconducting Nb wiring. We suggest that this contribution is a 'kinetic' inductance of the normal-metal film forming the resistor.

In a free electron model (Drude model), the complex conductivity of a metal in an electromagnetic field of angular frequency

ω is

$$\sigma(\omega) = \sigma_0 / (1 + i\omega\tau), \quad (10)$$

where τ is electron momentum relaxation (elastic scattering) time, $\sigma_0 = \frac{ne^2\tau}{m}$ is the ordinary dc conductivity, n and m are the electron number density and electron effective mass, respectively; see e.g., [41]. Hence, for a thin film with thickness d and uniform current distribution, the imaginary part of the rf impedance $1/\sigma(\omega)$ is $\omega L = \sigma_0^{-1}\omega\tau l/(wd)$. The film inductance (Drude inductance) is simply

$$L = \left(\frac{l}{w}\right) R_{sq}\tau = N_{sq} \frac{1}{d} \frac{m}{ne^2} = N_{sq} L_{sq}, \quad (11)$$

where $L_{sq} = m/(ne^2d)$ and $R_{sq} = 1/(\sigma_0 d)$ is the film's sheet resistance at dc.

The origin of this inductance is in electron inertia. It is an exact analog of the kinetic inductance of superconducting films and depends in the same manner on m/n and d . This 'kinetic' inductance of normal-metal films is usually neglected because, in the typical thin films with short mean free path, $\omega\tau \ll 1$ and $\omega L_{sq} \ll R_{sq}$. For our 40-nm Mo film, we can estimate the elastic scattering time as $\tau = 0.57$ ps, using $L_{sq} = 1.132$ pH/sq from the fit in Fig. 13(b), $R_{sq} = 2 \Omega/\text{sq}$, and (11). Therefore, the condition of validity of (10), $\omega\tau \ll 1$, is satisfied at all frequencies of interest up to $\omega_L \approx 1.8 \cdot 10^{12}$ rad/s. The uniform rf current distribution is also realized because $d \ll \delta$, where δ is the skin depth. Thus, the LRC resonance in shunted JJs gives a simple method of extracting elastic scattering time τ in various thin films by incorporating them as a JJ shunt resistor or its part.

D. Scaling With Junction Size, J_c , and Sheet Resistance

The scaling with junction size is very simple and follows from (8). The required value of β_C defines the shunt resistance. If the subgap resistance can be neglected, the shunt resistance in the first approximation is $R_{sh} \approx \frac{1}{A_J} \left(\frac{\beta_C \Phi_0}{2\pi J_c C_s} \right)^{1/2}$ and equals $N_{sq} R_{sq}$. Similarly, the shunt inductance is $L_{sh} = L_{sq} N_{sq} + L_g$. Then, in the first approximation, ω_r does not depend on the junction size and depends only on elastic scattering time and the tunnel barrier properties, and is given by

$$\omega_r^2 = \frac{1}{\tau} \left(\frac{2\pi J_c}{\beta_C \Phi_0 C_s} \right)^{1/2} - \frac{1}{\tau^2} \quad (12)$$

This result is consistent with Fig. 4 showing that the upward bend in the I - V characteristics, indicating the resonance, occurs almost at the same voltage, see also Table I. In reality, V_r slightly decreases because, at small resistor values (large JJ sizes), the inductance of the resistor wiring becomes comparable to the inductance of the resistor film. As a result, the quality factor increases, making V_r and V_m closer to each other and the resonance features more pronounced in larger JJs, as can be seen in Fig. 4.

Increasing β_C to the typical for SFQ circuits value $\beta_C \approx 1$ by increasing the resistor length l , while keeping the same sheet resistance $R_{sq} = 2 \Omega/\text{sq}$, increases proportionally L_{sh} and decreases the resonance frequency to $V_r = 0.25$ mV, creating a

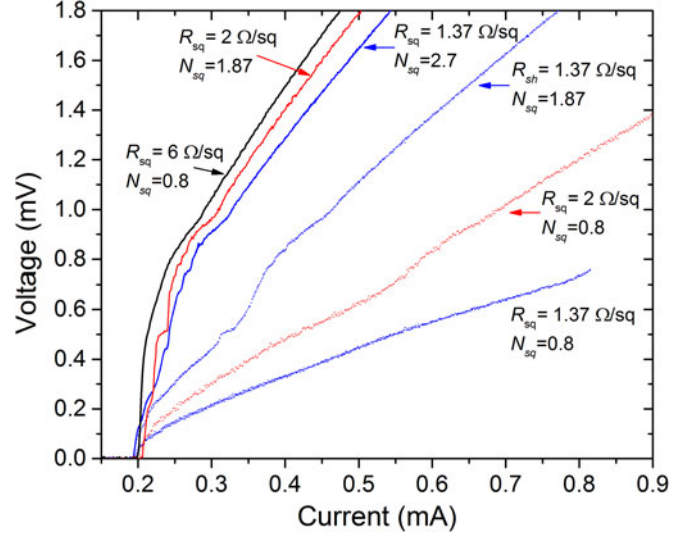


Fig. 14. I - V characteristics of 1.6- μm diameter junctions shunted by different shunt resistors obtained by increasing N_{sq} and the sheet resistances R_{sq} : 1.37 Ω/sq (Mo, blue curves); 2.0 Ω/sq (Mo, red curves); 6 Ω/sq (MoN_x, black solid curve). The resonance features were not observed for MoN_x shunts.

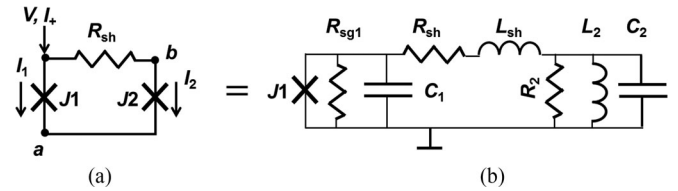


Fig. 15. A test circuit used for extracting Josephson plasma resonance frequency from I - V characteristics (a) and its circuit diagram (b). The dc bias current was extracted from port "a" making J1 and J2 connected in parallel at dc. Extracting the current from port "b" would make them connected in series as in [42] and [43]. In both cases, the junctions are connected identically for the rf current generated by J1. In (b) J2 is replaced by a parallel combination of junction capacitance C_2 , Josephson inductance L_2 , and subgap resistance R_2 ; L_{sh} is the shunt inductance.

rich structure in I - V characteristics as shown in Figs. 5 and 14. Since β_C grows as l^2 and β_L as l , the resonance disappears with increasing the length further.

The most efficient way of suppressing the resonance is to increase the sheet resistance R_{sq} of the resistor film by increasing electron scattering rate τ^{-1} , while keeping the same film thickness and N_{sq} , because this increases the β_C without increasing the β_L . We increased R_{sq} by a factor of 3x, from 2 Ω/sq to 6 Ω/sq , by replacing Mo film with MoN_x (nitrogen-doped molybdenum) films of the same thickness [5]. Nitrogen doping mainly increases the impurity scattering rate τ^{-1} .

A comparison of the I - V characteristics of two junctions, one shunted by Mo resistor ($R_{sq} = 2 \Omega/\text{sq}$, $\beta_C = 0.22$) and another one by MoN_x resistor ($R_{sq} = 6 \Omega/\text{sq}$, $\beta_C = 2$) with the absolutely identical design is shown in Fig. 14. It is clear that the resonance feature disappeared from the I - V curve of the MoN_x-shunted JJ, because MoN_x provides higher shunt resistance at the same shunt inductance. In other words, the LRC resonance associated with the 'kinetic' induc-

tance of the shunt resistor can be completely damped by using any resistor material having a shorter electron scattering time than the $R_{sh}C$ time constant of the junction capacitor, $R_{sh}C \geq \tau$.

In almost all shunted junctions studied, L_{sh} is about a few pH. This parasitic inductance in some cases may become comparable with the signal inductances in SFQ cells. Its existence may affect timing characteristics and high-frequency dynamics of the individual cells and margins of SFQ circuits. Therefore, it may be important to include L_{sh} into the junction models for SFQ circuit simulation.

IV. JOSEPHSON PLASMA RESONANCE FREQUENCY

Knowledge of the Josephson plasma resonance frequency (voltage)

$$\omega_p = \left(\frac{2\pi I_c}{C\Phi_0} \right)^{1/2} = \left(\frac{2\pi J_c}{C_s\Phi_0} \right)^{1/2}, \quad (13a)$$

$$V_p = (\Phi_0/2\pi)\omega_p, \quad (13b)$$

corresponding to the internal LC resonance between the junction capacitance C and the Josephson inductance $L_J = \Phi_0/(2\pi I_c)$, is important for circuit design in order to correctly determine the critical damping $\omega_p\tau_C = 1$ and the characteristic voltage $V_c = I_c R_s$ corresponding to the critical damping $V_c = V_p$, where $\tau_C = R_s C$.

One of the methods of measuring ω_p and the junction capacitance is to excite the Josephson plasma resonance in a large-area junction, $J2$ using a small junction $J1$ as a source of electromagnetic oscillations. A circuit where $J2$ is coupled via resistor R_{sh} was described in [42] and shown in Fig. 15. The $J2$ impedance is very low at all frequencies except near the ω_p because $J2$ is either shorted by its inductance or its capacitance. At ω_p , the $J2$ impedance becomes real, that should result in a feature in the dc I - V characteristic of $J1$. If both the rf and dc currents flowing through $J2$ are small, an unperturbed value of (13) can be extracted. This requires the critical current of $J1$, I_{c1} be much smaller than the critical current of $J2$, I_{c2} . The circuit was used in [15], [42], [43] for extracting ω_p and the junction specific capacitance C_s . Below, we will give a more detailed analysis of the circuit in order to extract ω_p and C_s in high- J_c junctions.

First of all, there are two possible ways of connecting $J1$ and $J2$ with respect to the dc bias current as shown schematically in Fig. 15(a), using either port “a” or “b” as the circuit ground. The junctions are connected in parallel if the bias current is taken out of the port “a”; whereas they are connected in series if the current is taken out of the port “b” as in [42], [43]. However, the rf connection is identical in both cases. For Josephson oscillations generated by $J1$, the serially connected shunt resistor and $J2$ are in parallel to $J1$.

As a part of our standard process control monitors, we used the circuit with port “a” in order to reduce the current, I_2 flowing through $J2$ and its effect on ω_p . The circuit is almost identical to the one shown in Figs. 10 and 11 with the only difference being that I5 via is replaced by the large-area junction $J2$. We used

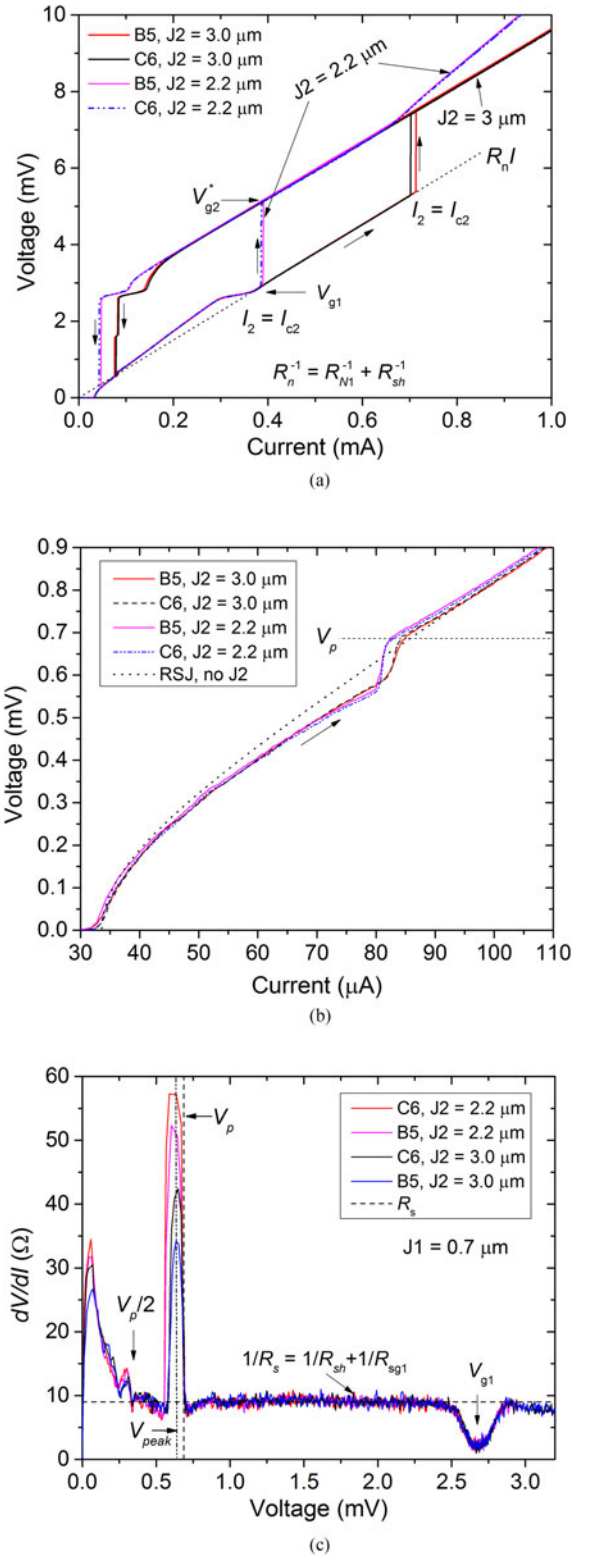


Fig. 16. (a) I - V characteristics and dV/dI for the test structures in Fig. 15 with $J1 = 0.7 \mu\text{m}$, $R_{sh} = 9.7 \Omega$, and $J2 = 2.2 \mu\text{m}$ and $3.0 \mu\text{m}$ at two locations, B5 and C6, on the same wafer. (b) Low-voltage part of the I - V characteristics showing the region of the Josephson plasma resonance in $J2$; the dotted line shows the RSJ model dependence without $J2$. (c) Differential resistance dV/dI versus voltage dependencies showing the peak at a voltage corresponding to a resonance of a parallel type in the structure; the curves from top to bottom correspond to the test structures in the legend.

TABLE II
PARAMETERS OF THE CIRCUIT IN FIG. 15 AT $J_c = 0.1 \text{ mA}/\mu\text{m}^2$

J1 dia (μm)	$R_{\text{sh}} (\Omega)$	$L_{\text{sh}} (\text{pH})$	J2 dia (μm)	dV/dI peak voltage (mV)	V_{peak} Simulated (mV)	$C_s (\text{fF}/\mu\text{m}^2)$
0.7	9.7	6.6	2.2	0.62	0.619	70
0.7	9.7	6.6	3.0	0.64	0.640	70
1.0	5.0	3.5	4.0	0.62	0.630	70
1.0	5.0	3.5	5.0	0.615	0.641	70

two sizes of the ‘active’ junction $J1$, $0.7 \mu\text{m}$ and $1.0 \mu\text{m}$, and varied the diameter of the ‘passive’ junction $J2$ from $2.2 \mu\text{m}$ to $6 \mu\text{m}$.

The typical I - V characteristics of $J1$ in a wide range of currents are shown in Fig. 16(a). The structure works as follows. At small currents $I \leq I_{c1}$, the current flows only through junction $J1$ because it is in the superconducting state. Above the critical current I_{c1} , a part of the total current, I_2 starts flowing through the shunt resistor and junction $J2$, providing a frequency-dependent damping of Josephson oscillations in $J1$. Junction $J2$ is in the superconducting state because its critical current $I_{c2} \gg I_{c1}$. Eventually, the gap voltage V_{g1} is reached across $J1$, above which the I - V characteristics becomes linear with the slope $R_n^{-1} = R_{N1}^{-1} + R_{\text{sh}}^{-1}$, shown by a dotted line in Fig. 16(a), where R_{N1} is the normal-state resistance of $J1$. When I_2 reaches I_{c2} , junction $J2$ switches to its gap voltage, because $J2$ is strongly underdamped. The total voltage across $J1$ jumps up by $V_{g2} = V_{g1} \approx 2.7 \text{ mV}$. This is shown by the vertical arrows in Fig. 16(a) for the two sizes of $J2$ used. The total dc voltage across the structure after the switching is $V_{g2}^* = R_n I + V_{g2}$, as shown in Fig. 16(a) for $J2 = 2.2 \mu\text{m}$. Then, on the gap voltage branch of $J2$, the voltage across $J2$ is constant and equal V_{g2} (the junction has nearly zero differential resistance). So, the I - V characteristic remains linear with the same slope until the current through $J2$ becomes higher than the ‘knee’ current. Above this point, the slope of the I - V characteristic increases and becomes $V(I) = I[R_{N1}^{-1} + (R_{\text{sh}} + R_{N2})^{-1}]^{-1}$. This increase in the slope is clearly seen for $J2 = 2.2 \mu\text{m}$ above about 0.67 mA . Upon decreasing the current, there is a big hysteresis because $J2$ remains in the resistive state and switches back into the S -state at a lower current as shown in Fig. 16(a). So both I_{c1} and I_{c2} , and other parameters of the test structure can be found from Fig. 16(a).

A zoom in the region of voltages of interest, Fig. 16(b), shows a sharp increase in dc voltage across $J1$ in the voltage range near the expected resonance voltage $V_p = \omega_p \Phi_0 / (2\pi)$. Accordingly, the derivative dV/dI versus V displays a sharp peak at a voltage V_{peak} , corresponding to a resonance of a parallel type in the structure; see Fig. 16(c). In [42], this peak was associated with the Josephson plasma resonance in $J2$, and its position was used to extract C_s in [15], [42], [43].

However, a simple analysis shows that the resonance that really occurs in the circuit is a parallel RLC -type resonance discussed in III in connection with the shunt inductance. The only difference from Figs. 10 and 11 is that a superconducting

via $I5$ with nearly zero inductance is replaced by the junction $J2$ with a frequency-dependent inductance, as shown in Fig. 15(b).

The resonance is created by the ‘active’ junction capacitance C_1 in parallel with the impedance of the shunt- $J2$ branch, $Z(\omega)$

$$Z(\omega) = R_{\text{sh}} + i\omega L_{\text{sh}} + Z_2(\omega), \quad (14)$$

where $Z_2(\omega)$ is the equivalent impedance of $J2$,

$$Z_2(\omega) = R_2 \left[\frac{(\omega/\omega_L)^2}{(1 - \omega^2/\omega_p^2)^2 + \omega^2/\omega_L^2} + i \frac{(\omega/\omega_L)(1 - \omega^2/\omega_p^2)}{(1 - \omega^2/\omega_p^2)^2 + \omega^2/\omega_L^2} \right], \quad (15)$$

and $\omega_L = R_2/L_2$. The resonance frequency, ω_{res} is given by a solution of the equation

$$\omega C_1 Z(\omega) Z^*(\omega) - \text{Im}[Z(\omega)] = 0, \quad (16)$$

which is a generalized form of (3).

At $\omega < \omega_p$, the $J2$ impedance is inductive. It adds up to L_{sh} , thus decreasing the circuit resonance frequency ω_{res} with respect to ω_r in (3) and increasing the Q -factor. Hence, the dV/dI peak voltage, given by (6) for the no- $J2$ case ($L_2 = 0$), shifts down from the values given in Table I to the lower values shown in Table II and in Fig. 16(c). So, essentially, the I - V features in Fig. 16(b) and (c) are the same as those in Fig. 4, only shifted and sharpened by the presence of the $J2$ inductance.

For $0.1 \text{ mA}/\mu\text{m}^2$ JJs, ω_L (in voltage units) is $2\pi I_c R_{\text{sg}}/\Phi_0 \sim 18 \text{ mV}$. So, the Q -factor of the $J2$ resonator $Q_2 = \omega_L/\omega_p$ is high, $Q_2 \sim 26$. Therefore, at all frequencies (voltages) $V < V_p$, except in the immediate vicinity of V_p , $\text{Im}(Z) \approx \omega(L_{\text{sh}} + L_2)$. The solution of (16) can then be approximated by (3) and V_{peak} value by (6) with the effective inductance $L_{\text{sh}} + L_2$.

In the limit of negligible R_{sh} and L_{sh} , the C_1 capacitor is in parallel with $J2$, making the resonance at $\omega_{\text{res}} = [L_2(C_2 + C_1)]^{-1/2}$ and lowering the resonance voltage to $V_{\text{res}} = V_p/(1 + A_1/A_2)^{1/2}$. At intermediate values, a circuit simulator can be used to extract C_2 and ω_p from the peak voltage V_{peak} , using L_{sh} values of the corresponding shunt resistors from Table I, and doing the measurements at different sizes of $J2$ and for junctions with different J_c values. At voltages higher than V_p , the $J2$ impedance becomes capacitive and resonance with C_1 is no longer possible. In all the cases, the V_p corresponds closely to the minimum in dV/dI on the right side of the peak, as shown in Fig. 16(b) and (c). We note that without knowledge of the value of the coupling (shunt) inductance connected to $J2$, the procedure of extracting ω_p used in [42], [43] becomes somewhat ambiguous because it involves too many fitting parameters.

The data in Fig. 16 for $J_c = 0.1 \text{ mA}/\mu\text{m}^2$ serve as a self-consistency check. In III, we used $C_s = 70 \text{ fF}/\mu\text{m}^2$ to extract L_{sh} . Now we use this L_{sh} and the same C_s , corresponding to $V_p = 0.686 \text{ mV}$, to calculate the V_{peak} in a different structure. The agreement with the measurements is excellent as shown in Table II. However, in all cases, V_{peak} is noticeably lower than V_p . Therefore, using V_{peak} value instead of V_p to calculate C_s from (13) would noticeably overestimate the junction specific capacitance. As a consistency check we have done the

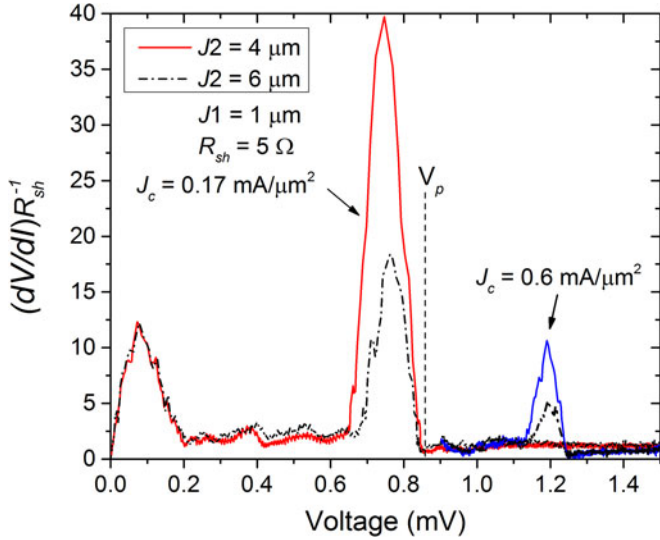


Fig. 17. Differential resistance dV/dI versus V dependence for the test structures in Fig. 15 with $J_1 = 1.0 \mu\text{m}$ and $J_2 = 4.0 \mu\text{m}$ (solid curves) and $6.0 \mu\text{m}$ (dash-dotted curves) at $J_c \approx 0.2 \text{ mA}/\mu\text{m}^2$ (left peaks) and $0.6 \text{ mA}/\mu\text{m}^2$ (right peaks).

measurements for different sizes of J_2 . The results are shown in Table II. As can be seen, the dV/dI peak voltage very weakly depends on the J_2 size, as expected.

In general, the plasma resonance frequency depends on the dc current through the junction J_2 as $\omega_p^2 = (1 - I_2^2/I_{c2}^2)^{1/2} \omega_{p0}^2$, so the plasma resonance voltage shifts to a lower voltage $V_p = V_{p0} [1 - (I_2/I_{c2})^2]^{1/4}$. At the resonance voltage in our structure $I_2 = V_{\text{peak}}/R_s \sim 0.1 I_{c2}$, and this correction is only about 1%. We will neglect this tiny difference.

After this analysis, we turn to the higher- J_c junctions, using the same circuit with exactly the same R_{sh} and L_{sh} . With increasing J_c , the I - V characteristics of the junctions and dV/dI acquire additional features due subharmonic pumping of the resonance. The typical voltage dependences of the normalized differential resistance $(dV/dI)/R_{sh}$ for the test structure in Fig. 15 with high- J_c junctions are shown in Fig. 17.

The maximum differential resistance in Figs. 16(c) and 17 decreases with increasing the J_2 size and/or increasing J_c because the Q -factor decreases as a result of decreasing R_2 and the junction inductance L_2 . Also, as ω_p increases with J_c , microwave losses in the structure also increase with frequency, reducing the Q -factor. At $V_p \geq \Delta/e \sim 1.35 \text{ mV}$, the frequency of Josephson oscillations in the structures $2eV_p/h$ becomes larger than the absorption threshold frequency $2\Delta/h$. The dissipation strongly increases due to pair breaking, damping the plasma oscillations. This sets the maximum observable Josephson plasma resonance frequency.

The voltage of the maximum in $(dV/dI)/R_{sh}$ and the extracted values of V_p are shown in Fig. 18 for the entire range of J_c s studied. Both quantities increase with J_c but much slower than $J_c^{1/2}$ following from (13b), apparently because of increasing specific capacitance of the junctions. The V_p values obtained are consistent with the data on the maximum clock frequency observed in RSFQ T-flip flops [1], [9]. The maximum value of

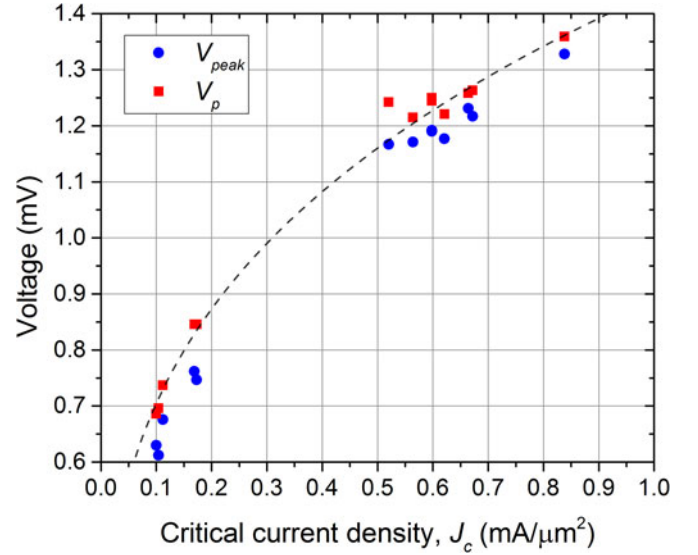


Fig. 18. The voltage V_{peak} of the peak in differential resistance dV/dI of the active junction J_1 coupled to a passive junction J_2 through the shunt resistor (blue circles) and the extracted values of the Josephson plasma resonance voltage (13b) in Nb/AlO_x-Al/Nb Josephson junctions with different critical current densities, J_c (red squares). The dashed line is to guide the eye.

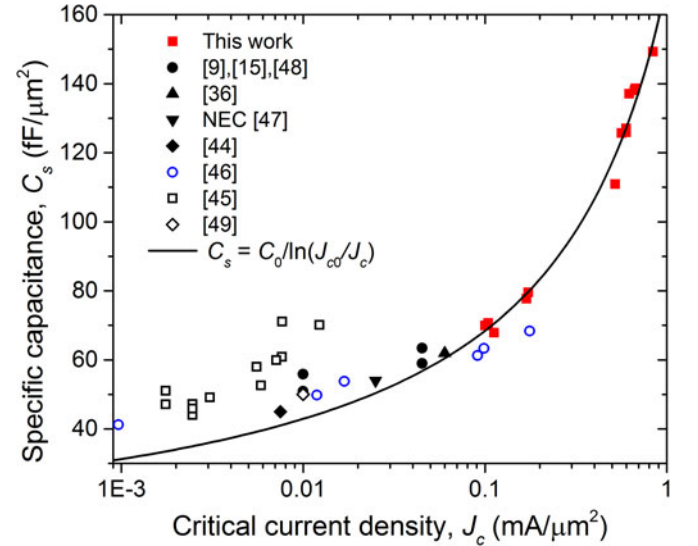


Fig. 19. The specific capacitance of Nb/AlO_x-Al/Nb junctions with different critical current density extracted from the I - V characteristics of the test structures in Fig. 15. Historic data are also included. The solid line is the best fit of our data by the dependence $C_s = C_0 / \ln(J_{c0}/J_c)$ expected from the simplest model (17); see text.

the V_p observed, $\sim 1.35 \text{ mV}$ is consistent with the pair-breaking criterion above.

The junction specific capacitance calculated using V_p and (13) is shown in Fig. 19. In the simplest model, $J_c = J_{c0} \exp(-\alpha d)$ and $C_s = \epsilon \epsilon_0 / d$, where d is the tunnel barrier thickness.

Hence, the specific capacitance changes inversely with the logarithm of J_c as

$$C_s = C_0 / \ln(J_{c0}/J_c) \quad (17)$$

The solid curve in Fig. 19 shows the best fit of this dependence to our data, giving $C_0 = 265 \pm 15 \text{ fF}/\mu\text{m}^2$ and $J_{c0} = 4.8 \pm$

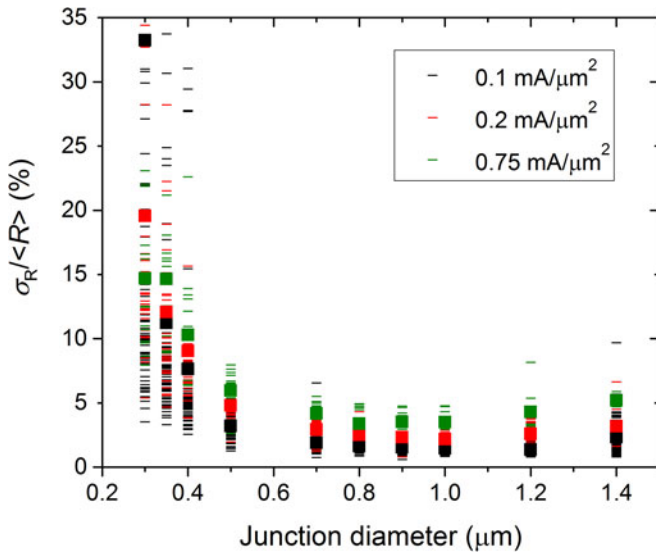


Fig. 20. Normalized standard deviation of the normal-state resistance of nominally identical junctions tested on the fabricated wafers. Each data point (dash) corresponds to the data obtained for junctions of each size located on one die. Nine die locations per 200-mm wafer were measured, 990 JJs of one given junction size per wafer. The data on 16 wafers are included: eight wafers with $J_c = 0.1 \text{ mA}/\mu\text{m}^2$ (black dashes), four wafers with $J_c = 0.2 \text{ mA}/\mu\text{m}^2$ (red dashes), and four wafers with $J_c = 0.75 \text{ mA}/\mu\text{m}^2$ (green dashes).

$0.5 \text{ mA}/\mu\text{m}^2$. For completeness we also included in Fig. 19 the literature data on the specific capacitance [9], [15], [44]–[49], obtained by using *LC* resonance in SQUIDs and/or a zero-field resonance, Fiske step in large junctions.

V. PARAMETER SPREADS OF HIGH- J_c JUNCTIONS

The width of the distribution of the critical currents of the nominally identical junctions or, equivalently, the width of the distribution of their conductances is an important parameter characterizing the state of the fabrication technology and the maximum complexity of digital circuits [3]. As was shown in our previous work [4], these junction parameter spreads can be well described by a normal (Gaussian) distribution. The normalized (to the mean value) standard deviation of the junction normal-state resistance $\sigma_R / \langle R \rangle$ is shown in Fig. 20 for junctions with sizes from $0.3 \mu\text{m}$ to $1.4 \mu\text{m}$ in diameter at three values of J_c corresponding to SFQ5ee ($0.1 \text{ mA}/\mu\text{m}^2$) and SFQ5hs ($0.2 \text{ mA}/\mu\text{m}^2$) process node targets, and a yet higher value $J_c = 0.75 \text{ mA}/\mu\text{m}^2$, corresponding to nearly the end of the J_c range studied.

We note that the minimum junction diameter which can be printed using our deep-UV photolithography tool is $0.25 \mu\text{m}$ [4], and the minimum size allowed by the circuit design rules for our processes is $0.7 \mu\text{m}$. It can be seen that the spreads of the junction resistances is nearly the same in junctions with $J_c = 0.1 \text{ mA}/\mu\text{m}^2$ and $J_c = 0.2 \text{ mA}/\mu\text{m}^2$, and only slightly higher in the highest J_c case. The size-dependence is consistent with the model given in [4]. Some locations show nearly the same parameter spreads for all three current densities, including the highest one, indicating that, in some cases, the junction

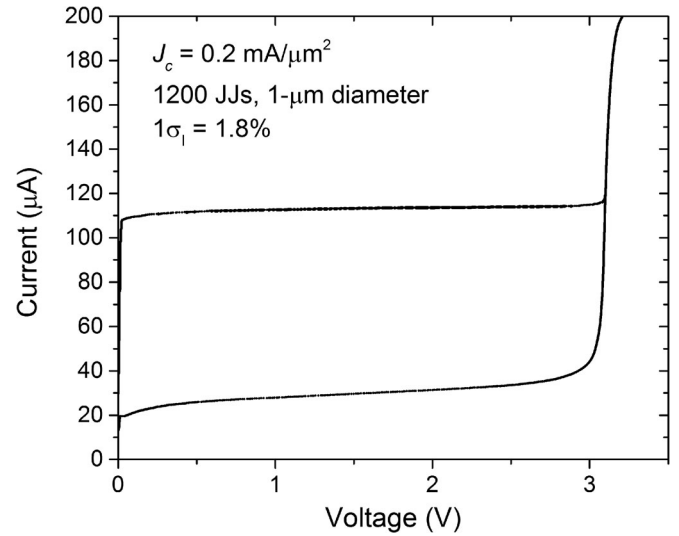


Fig. 21. I - V characteristics of a series array of 1200 junctions with design diameter of $1 \mu\text{m}$ and $J_c = 0.2 \text{ mA}/\mu\text{m}^2$. The length of the array is about 4 mm . It characterizes the typical distribution of critical currents of the nominally identical junctions within one chip, giving a standard deviation of the critical current distribution of about 1.8%.

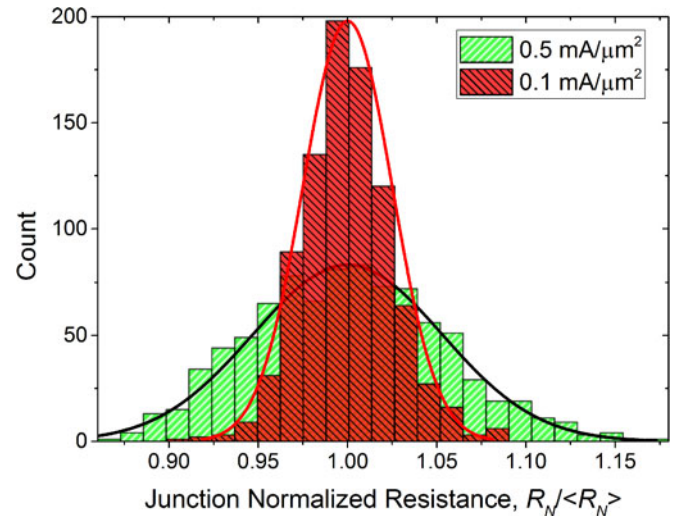


Fig. 22. Histograms of the resistance of $0.7\text{-}\mu\text{m}$ -diameter junctions comparing two 200-mm wafers, with $J_c = 0.1 \text{ mA}/\mu\text{m}^2$ and $J_c = 0.5 \text{ mA}/\mu\text{m}^2$. In each case 880 JJs were measured, locations for the junctions were identical in both cases. Junction resistance was normalized to the respective wafer mean value $\langle R_N \rangle$. Solid lines show the fits by Gaussian distributions with $\sigma_R = 2.5\%$ and 5.3% , respectively, for $0.1\text{-mA}/\mu\text{m}^2$ and $0.5\text{-mA}/\mu\text{m}^2$ junctions.

spreads are not yet limited by the barrier properties and can be further reduced by perfecting the fabrication process.

The results of the I_c measurement at 4.2 K using an array of 1200 junctions with $1.0 \mu\text{m}$ design diameter and $J_c = 0.2 \text{ mA}/\mu\text{m}^2$ are shown in Fig. 21. The results are fully consistent with the spreads obtained from the room temperature junction resistance measurements, as was shown in [4]. This once again justifies the use of room- T junction resistance measurements for characterizing junction parameter spreads on the wafer scale.

A more detailed comparison of $0.1\text{-mA}/\mu\text{m}^2$ and $0.5\text{-mA}/\mu\text{m}^2$ junctions is given in Fig. 22 showing the aggregate distributions of room temperature resistance, R_N of $0.7\text{-}\mu\text{m}$ junctions on two 200-mm wafers fabricated with these current densities. Junction resistances were normalized to the wafer mean value $\langle R_N \rangle$. Locations of the junctions and their number were identical in both cases. Solid lines show fits to the normal distribution giving normalized standard deviations $\sigma_R = 2.5\%$ and 5.3% , respectively, for $0.1\text{-mA}/\mu\text{m}^2$ and $0.5\text{-mA}/\mu\text{m}^2$ junctions. Since these wafers were processed identically, the contribution of junction area fluctuations [4] should be identical for both current densities. Therefore, the wider statistical distributions of the higher- J_c junctions can only be attributed to a higher contribution of fluctuations in AlO_x barrier transparency as the barrier gets thinner.

VI. CONCLUSION

We have studied the fabrication parameters and electrical properties of $\text{Nb}/\text{AlO}_x\text{-Al}/\text{Nb}$ junctions, unshunted and with resistive shunting, in the range of Josephson critical current densities from $0.1\text{ mA}/\mu\text{m}^2$ to $\sim 1\text{ mA}/\mu\text{m}^2$ for the use in our high- J_c technology nodes. Junctions with $J_c = 0.1\text{ mA}/\mu\text{m}^2$ and $0.2\text{ mA}/\mu\text{m}^2$ are highly hysteretic and require resistive shunting for applications in SFQ digital circuits. In this respect, design rules for our new $0.2\text{-mA}/\mu\text{m}^2$ process SFQ5hs are identical to our $0.1\text{-mA}/\mu\text{m}^2$ process SFQ5ee.

We have determined the inductance associated with the resistive shunts made of thin Mo and MoN_x films. We have found that the main part of the shunt inductance is the kinetic (Drude) inductance of the normal-metal film. For 40-nm Mo films this inductance is $1.13\text{ pH}/\text{sq}$; it is about a factor of 3 lower for 40-nm MoN_x films. The shunt inductance creates an *LRC* resonator with the junction capacitance if $\beta_L \geq \beta_C$, inducing multiple features affecting *I-V* characteristics of the shunted JJs and their switching dynamics. Although the resonance in the shunt *LRC* circuit can be easily suppressed by increasing the β_C parameter of the shunt and using higher sheet resistance MoN_x resistors, the shunt inductance remains and may significantly affect switching dynamics and timing of SFQ logic cells. We believe it should be taken into account in SFQ circuit design if accurate knowledge of the high-frequency dynamics and timing parameters is important.

Using a similar *LRC* resonance involving an additional “passive” junction *J2* replacing the via between the shunt resistor of junction *J1* and the bottom electrode of *J1*, we evaluated the Josephson plasma resonance frequency, junction characteristic voltage corresponding to the critical damping, and junction specific capacitance in the range of critical current densities from $0.1\text{ mA}/\mu\text{m}^2$ to about $1\text{ mA}/\mu\text{m}^2$. Our data are well described by the dependence $C_s = C_0/\ln(J_{c0}/J_c)$ following from the simplest model, and give the fitting parameters $C_0 = 265 \pm 15\text{ fF}/\mu\text{m}^2$ and $J_{c0} = 4.8 \pm 0.5\text{ mA}/\mu\text{m}^2$. This dependence also well describes the available specific capacitance data down to $J_c \approx 1\text{ mA}/\mu\text{m}^2$, thus covering the J_c range spanning about 3 orders in magnitude.

We compared the statistical distributions of the junctions of various sizes relevant to SFQ digital circuits and $J_c = 0.1, 0.2, 0.5$, and $0.75\text{ mA}/\mu\text{m}^2$. For circular junctions larger than $0.7\text{ }\mu\text{m}$ in diameter, parameter spreads of $0.1\text{-mA}/\mu\text{m}^2$ and $0.2\text{-mA}/\mu\text{m}^2$ junctions are essentially the same. This makes it possible to use $0.2\text{-mA}/\mu\text{m}^2$ junctions in superconducting VLSI circuits. At J_c of $0.5\text{ mA}/\mu\text{m}^2$ and higher the distribution of junctions on-chip and on-wafer is presently a factor of two wider than for $0.1\text{-mA}/\mu\text{m}^2$ junctions. This still should allow for making complex high-speed digital circuits using $0.5\text{-mA}/\mu\text{m}^2$ junctions, likely with external shunts. The junction quality may further improve with process maturation.

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